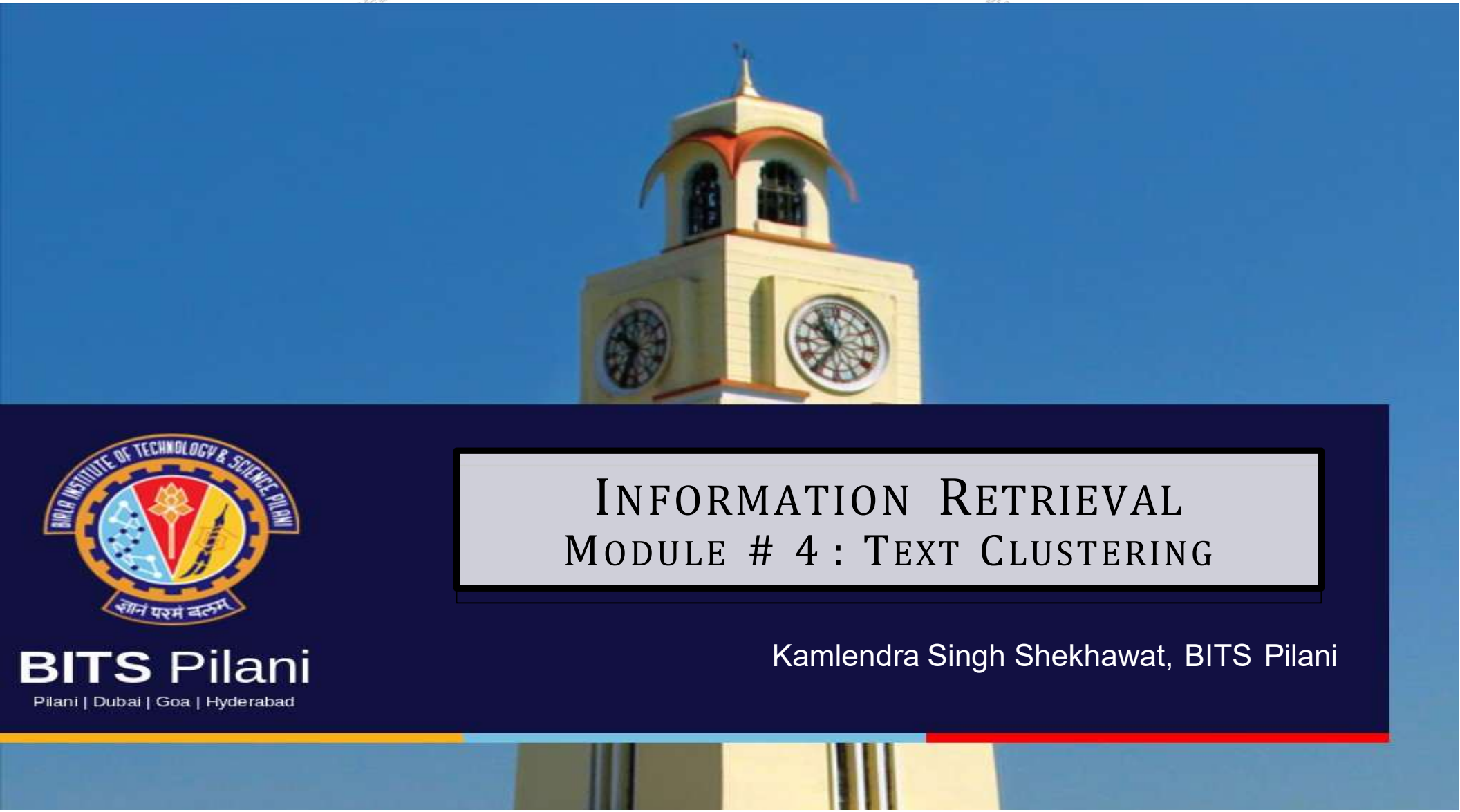

INFORMATION RETRIEVAL
MODULE # 4 : TEXT CLUSTERING

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Pilani | Dubai | Goa | Hyderabad

The instructor is gratefully acknowledging the authors who made their course materials freely available online.

This deck is prepared by Kamlendra Singh Shekhawat.

Work Integrated Learning Programmes

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1. Module 4
2. Document Clustering
3. Evaluating Clustering
4. Clustering Algorithms
5. Clustering Internet Entities

Module Topics....

- Document clustering
  - Motivations
  - Document representations
- Clustering algorithms
  - Flat clustering
  - Hierarchical clustering

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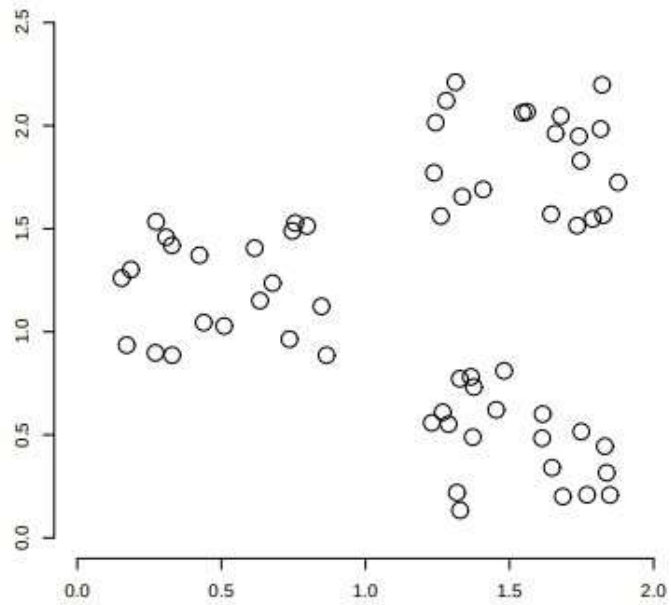
Clustering: Definition

- (Document) clustering is the process of grouping a set of documents into clusters of similar documents.
- Documents within a cluster should be similar.
- Documents from different clusters should be dissimilar.
- Clustering is the most common form of unsupervised learning.
- Unsupervised = there are no labeled or annotated data.

Classification vs. Clustering

- Classification: supervised learning
- Clustering: unsupervised learning
- Classification: Classes are human-defined and part of the input to the learning algorithm.
- Clustering: Clusters are inferred from the data without human input.
- However, there are many ways of influencing the outcome of clustering: number of clusters, similarity measure, representation of documents, . . .

Dataset with Clear Cluster Structure



Propose algorithm for finding the cluster structure in this example

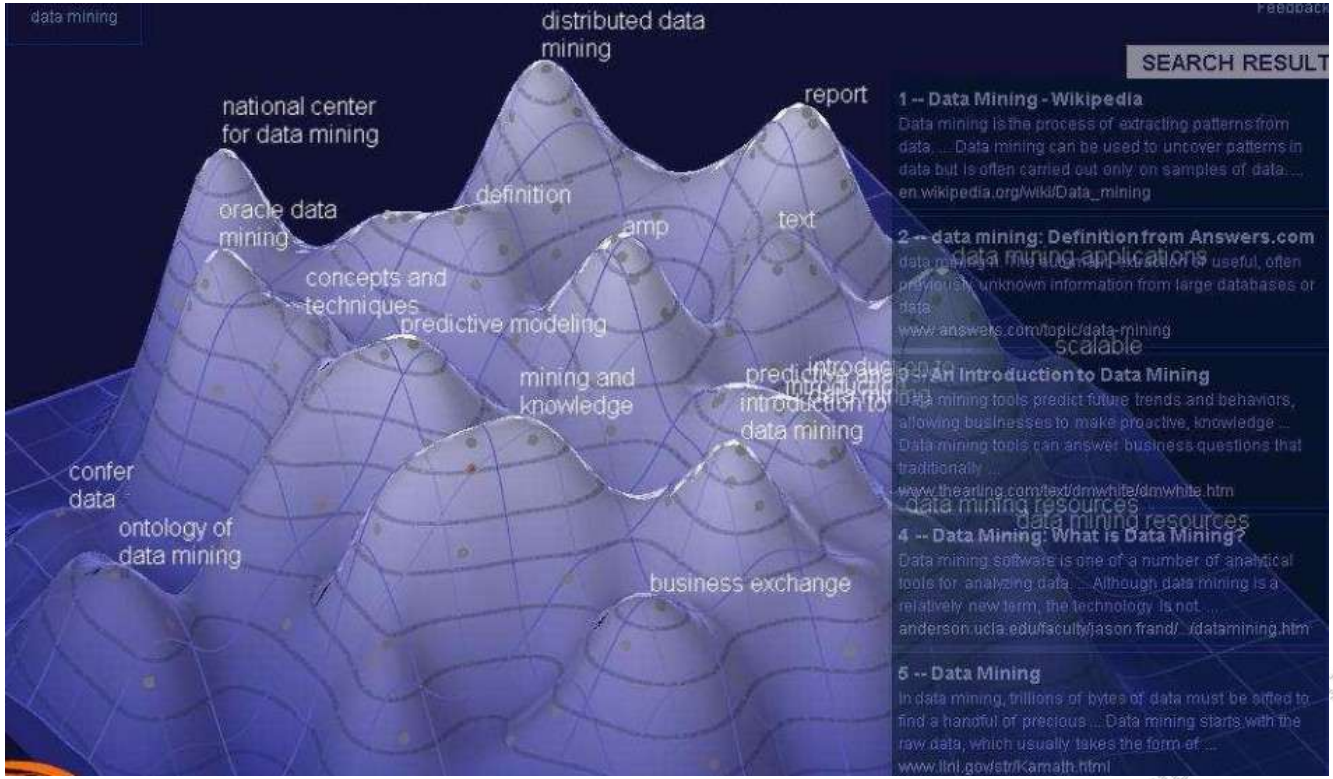
Applications of Clustering in IR

Application	What is clustered?	Benefit
Search result clustering	search results	more effective information presentation to user
Scatter-Gather	(subsets of) collection	alternative user interface: ”search without typing”
Collection clustering	collection	effective information presen-tation for exploratory browsing
Cluster-based retrieval	collection	higher efficiency: faster search

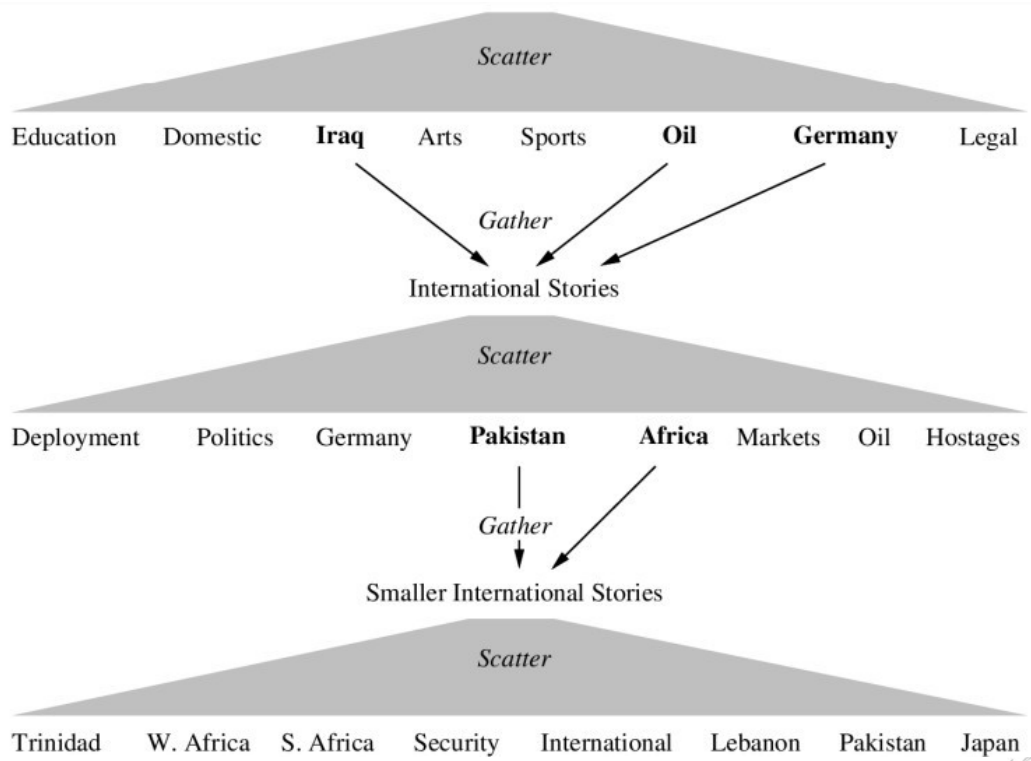
Search Result Clustering

► **Figure 16.2** Clustering of search results to improve recall. None of the top hits

Search Result Clustering



SCATTER-GATHER



GLOBAL NAVIGATION: YAHOO

YAHOO! DIRECTORY

Search: the Web | the Directory | this category

Search

Society and Culture

Directory > Society and Culture

SPONSOR RE

Culture
www.Dealtime.com Shop and save on Magazines.

CATEGORIES (What's This?)

Most Popular Society and Culture

- Crime (5453) NEW!
- Cultures and Groups (11025) NEW!
- Environment and Nature (8659) NEW!
- Families (1215)
- Food and Drink (9775) NEW!
- Holidays and Observances (3333)
- Issues and Causes (4842)
- Mythology and Folklore (984)
- People (16361)
- Relationships (595)
- Religion and Spirituality (37533)
- Sexuality (2812) NEW!

Additional Society and Culture Categories

- Advice (46)
- Chats and Forums (27)
- Cultural Policy (10)
- Death and Dying (394)
- Disabilities (1293)
- Employment and Work@
- Etiquette (54)
- Events (27)
- Fashion@
- Gender (21)
- Home and Garden (1080) NEW!
- Magazines (164)
- Museums and Exhibits (6062)
- Pets@
- Reunions (229)
- Social Organizations (338)
- Web Directories (6)
- Weddings (371)

SITE LISTINGS By Popularity | Alphabetical (What's This?)

Global Navigation: MESH (upper level)

MeSH Tree Structures - 2008

Return to Entry Page

1. ⊞ Anatomy [A]
2. ⊞ Organisms [B]
3. ⊟ Diseases [C]
   - Bacterial Infections and Mycoses [C01] +
   - Virus Diseases [C02] +
   - Parasitic Diseases [C03] +
   - Neoplasms [C04] +
   - Musculoskeletal Diseases [C05] +
   - Digestive System Diseases [C06] +
   - Stomatognathic Diseases [C07] +
   - Respiratory Tract Diseases [C08] +
   - Otorhinolaryngologic Diseases [C09] +
   - Nervous System Diseases [C10] +
   - Eye Diseases [C11] +
   - Male Urogenital Diseases [C12] +
   - Female Urogenital Diseases and Pregnancy Complications [C13] +
   - Cardiovascular Diseases [C14] +
   - Hemic and Lymphatic Diseases [C15] +
   - Congenital, Hereditary, and Neonatal Diseases and Abnormalities [C16] +
   - Skin and Connective Tissue Diseases [C17] +
   - Nutritional and Metabolic Diseases [C18] +
   - Endocrine System Diseases [C19] +
   - Immune System Diseases [C20] +
   - Disorders of Environmental Origin [C21] +
   - Animal Diseases [C22] +
   - Pathological Conditions, Signs and Symptoms [C23] +
4. ⊞ Chemicals and Drugs [D]
5. ⊞ Analytical, Diagnostic and Therapeutic Techniques and Equipment [E]
6. ⊞ Psychiatry and Psychology [F]
7. ⊞ Biological Sciences [G]
8. ⊞ Natural Sciences [H]
9. ⊞ Anthropology, Education, Sociology and Social Phenomena [I]
10. ⊞ Technology, Industry, Agriculture [J]
11. ⊞ Humanities [K]

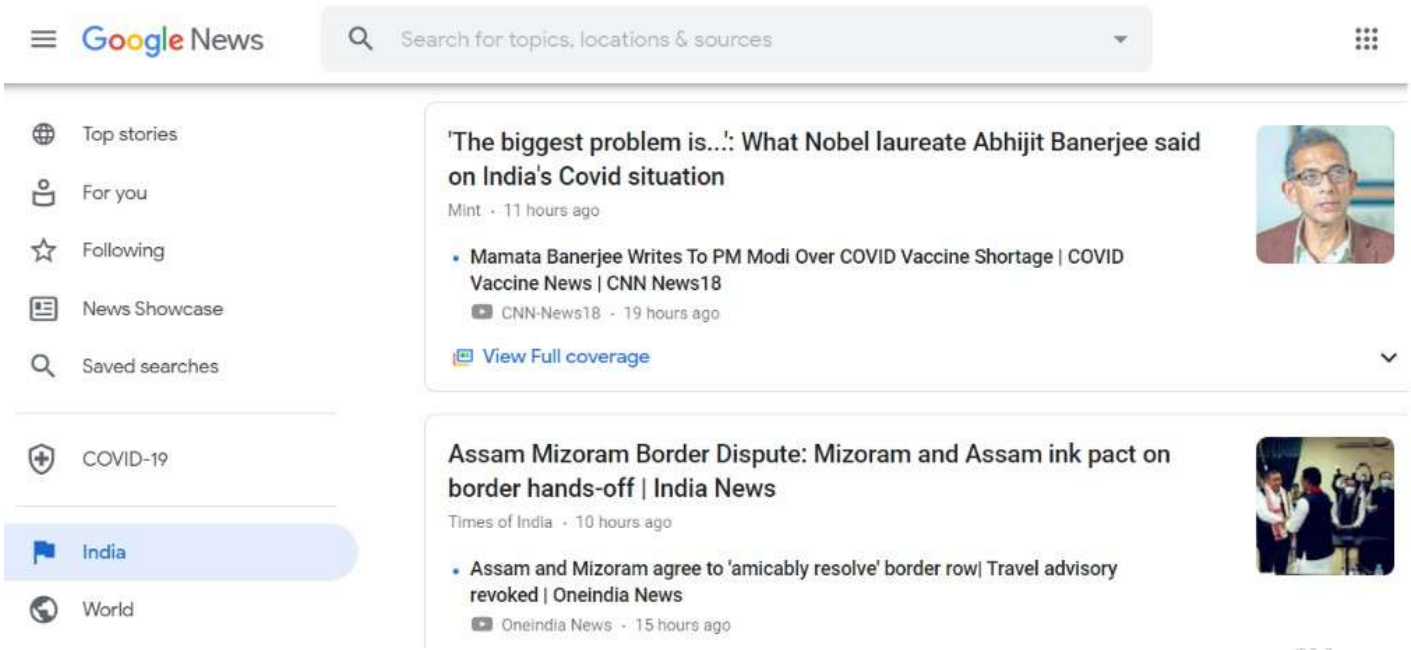
Global Navigation: MESH (lower level)

Neoplasms [C04]
- Cysts [C04.182] +
- Hamartoma [C04.445] +
- ▶ Neoplasms by Histologic Type [C04.557]
  - Histiocytic Disorders, Malignant [C04.557.227] +
  - Leukemia [C04.557.337] +
  - Lymphatic Vessel Tumors [C04.557.375] +
  - Lymphoma [C04.557.386] +
  - Neoplasms, Complex and Mixed [C04.557.435] +
  - Neoplasms, Connective and Soft Tissue [C04.557.450] +
  - Neoplasms, Germ Cell and Embryonal [C04.557.465] +
  - Neoplasms, Glandular and Epithelial [C04.557.470] +
  - Neoplasms, Gonadal Tissue [C04.557.475] +
  - Neoplasms, Nerve Tissue [C04.557.580] +
  - Neoplasms, Plasma Cell [C04.557.595] +
  - Neoplasms, Vascular Tissue [C04.557.645] +
  - Nevi and Melanomas [C04.557.665] +
  - Odontogenic Tumors [C04.557.695] +
- Neoplasms by Site [C04.588] +
- Neoplasms, Experimental [C04.619] +
- Neoplasms, Hormone-Dependent [C04.626]
- Neoplasms, Multiple Primary [C04.651] +
- Neoplasms, Post-Traumatic [C04.666]
- Neoplasms, Radiation-Induced [C04.682] +
- Neoplasms, Second Primary [C04.692]
- Neoplastic Processes [C04.697] +
- Neoplastic Syndromes, Hereditary [C04.700] +
- Paraneoplastic Syndromes [C04.730] +
- Precancerous Conditions [C04.834] +
- Pregnancy Complications, Neoplastic [C04.850] +
- Tumor Virus Infections [C04.925] +

Navigational Hierarchies: Manual vs. automatic creation

- Note: Yahoo/MESH are not examples of clustering.
- But they are well known examples for using a global hierarchy for navigation.
- Some examples for global navigation/exploration based on clustering:
  - Cartia
  - Google News

Global Clustering for Navigation: Google News



http://news.google.com

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Clustering for Improving Recall

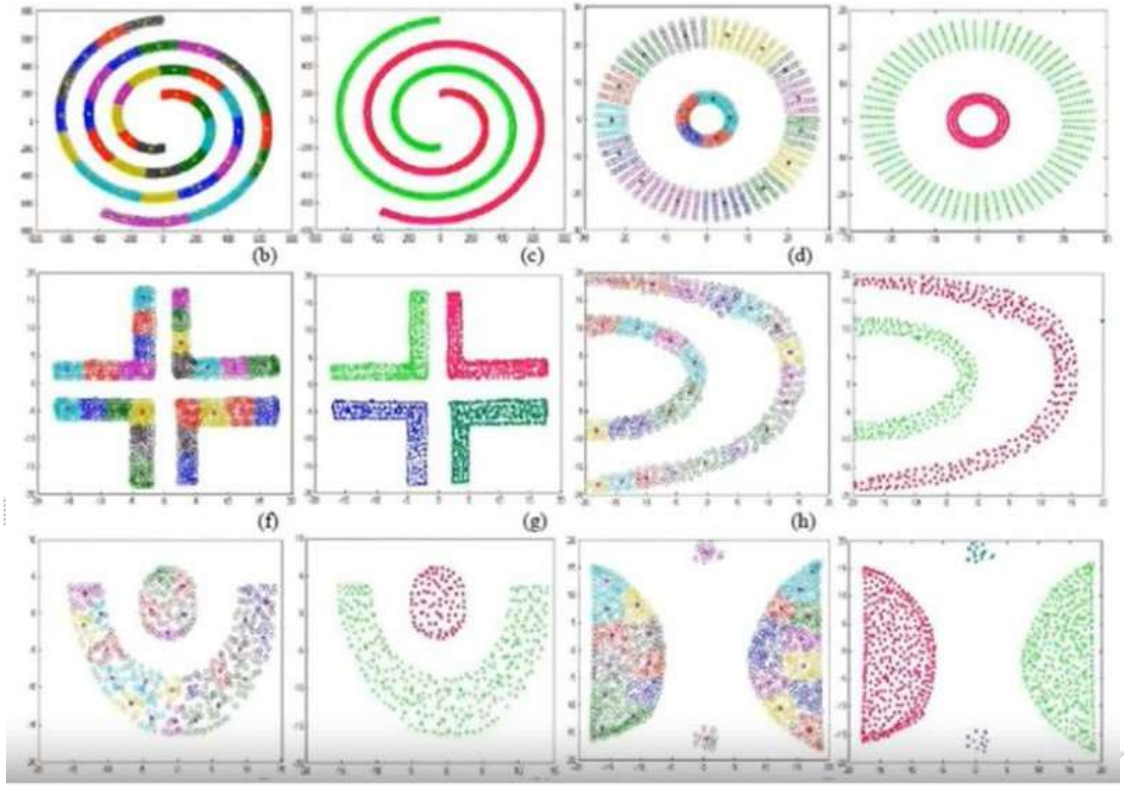
To improve search recall:

- Cluster docs in collection a priori
- When a query matches a doc d, also return other docs in the cluster containing d
- Hope: if we do this: the query ”car” will also return docs containing ”automobile”
- Because the clustering algorithm groups together docs containing ”car” with those containing ”automobile”.
- Both types of documents contain words like ”parts”, ”dealer”, ”mercedes”, ”road trip”.

Issues for Clustering

- Representation for clustering
  - Document representation
  - Vector space?
- Need a notion of similarity/distance
- How many clusters?
  - Fixed a priori?
  - Completely data driven?
- Avoid ”trivial” clusters - too large or small
- If a cluster’s too large, then for navigation purposes you’ve wasted an extra user click without whittling down the set of documents much.

Issues for Clustering



Number of Clusters

- General goal: put related docs in the same cluster, put unrelated docs in different clusters.
- How do we formalize this?
- The number of clusters should be appropriate for the data set we are clustering.
- Initially, we will assume the number of clusters K is given.
- Later: Semiautomatic methods for determining K
- Secondary goals in clustering
  - Avoid very small and very large clusters
  - Define clusters that are easy to explain to the user
  - Many others . . .

Hard vs. Soft clustering

- Hard clustering: Each document belongs to exactly one cluster.
  - More common and easier to do
- Soft clustering: A document can belong to more than one cluster.
  - Makes more sense for applications like creating browsable hierarchies
  - You may want to put sneakers in two clusters:
    - sports apparel
    - shoes
  - You can only do that with a soft clustering approach.

What is a Good Clustering?

- Internal criterion: A good clustering will produce high quality clusters in which:
  - the intra-class (that is, intra-cluster) similarity is high
  - the inter-class similarity is low
- The measured quality of a clustering depends on both the document representation and the similarity measure used

External Criteria for Clustering Quality

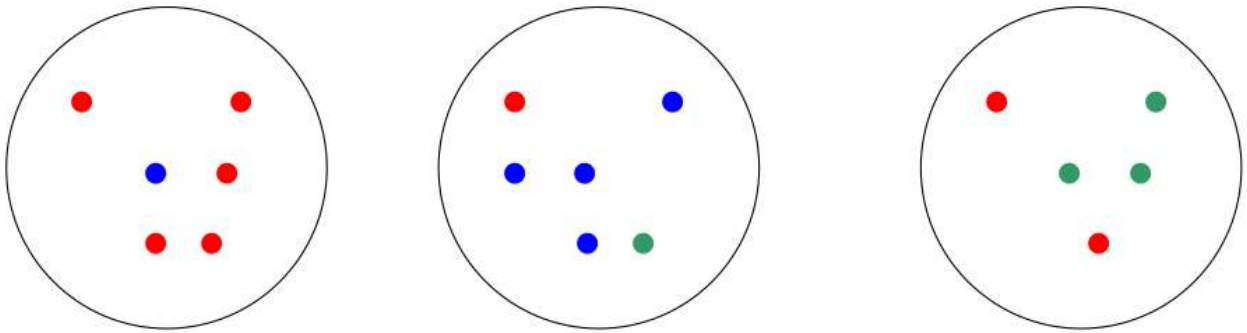
- Quality measured by its ability to discover some or all of the hidden patterns or latent classes in gold standard data
- Assesses a clustering with respect to ground truth . . . requires labeled data
- Assume documents with C gold standard classes, while our clustering algorithms produce K clusters, $\omega_1, \omega_2, \ldots, \omega_K$ with n_i members.

External Evaluation of Cluster Quality

- Simple measure: purity, the ratio between the dominant class in the cluster ω_i and the size of cluster ω_i
- Others are entropy of classes in clusters (or mutual information between classes and clusters)

$$Purity(\omega_i) = \frac{1}{n_i} \max_j |w_i \cap c_j|$$

Purity Example



$$Purity = \frac{1}{6}(5 + 4 + 3)$$

Rand Index

- Series of decisions, one for each of the $N(N-1)/2$ pairs of documents in the collection.
- Assign two documents to the same cluster if and only if they are similar.
  - True positive (TP) decision assigns two similar documents to the same cluster,
  - True negative (TN) decision assigns two dissimilar documents to different clusters.
- There are two types of errors we can commit.
  - A false positive (FP) decision assigns two dissimilar documents to the same cluster.
  - A false negative (FN) decision assigns two similar documents to different clusters.
- The Rand index (RI) measures the percentage of decisions that are correct.

Rand index and Cluster F-measure

$$P = \frac{TP}{TP + FP}$$

$$R = \frac{TP}{TP + FN}$$

$$RI = \frac{TP + TN}{TP + FP + FN + TN}$$

$$F - measure = \frac{TP}{TP + 0.5(FP + FN)}$$

Use a cluster F-measure, which is probably a better measure.

Rand Index Example

Number of point pairs	Same Cluster in clustering	Different Clusters in clustering
Same class in ground truth	20	24
Different classes in ground truth	20	72

$$RI = \frac{(20 + 72)}{(20 + 20 + 24 + 72)} \approx 0.68$$

$$F - measure = \frac{20}{20 + 0.5(24 + 20)} \approx 0.48$$

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Notion of Similarity/Distance

- Ideal: semantic similarity.
- Practical: term-statistical similarity (docs as vectors)
- Cosine similarity
- For many algorithms, easier to think in terms of a distance (rather than similarity) between docs.
- We will mostly speak of Euclidean distance
- But real implementations use cosine similarity

Clustering Algorithms

- Flat algorithms
  - Usually start with a random (partial) partitioning
  - Refine it iteratively
  - K means clustering
- Hierarchical algorithms
  - Bottom-up, agglomerative
  - Top-down, divisive

Partitioning Algorithms

- Partitioning method: Construct a partition of n documents into a set of K clusters
- Given: a set of documents and the number K
- Find: a partition of K clusters that optimizes the chosen partitioning criterion
- Globally optimal
  - Intractable for many objective functions
  - Ergo, exhaustively enumerate all partitions
- Effective heuristic method: K-means

K-Means

- Assumes documents are real-valued vectors.
- Clusters based on centroids (aka the center of gravity or mean) of points in a cluster, c:

$$\mu(c) = \frac{1}{|c|} \sum_{x \in c} x$$

- Reassignment of instances to clusters is based on distance to the current cluster centroids. (Or one can equivalently phrase it in terms of similarities)

K-Means Algorithm

1. Select K random docs $\{s_1, s_2, \ldots s_K\}$ as seeds.
2. Until clustering converges (or other stopping criterion):
   1. For each doc d_i:
      2. Assign d_i to the cluster c_j such that dist(x_i, s_j) is minimal.
   2. (Next, update the seeds to the centroid of each cluster)
   3. For each cluster c_j
      2. $s_j = \mu(c_j)$

K-Means Algorithm

1. Initialize cluster centers
2. Assign observations to the closest cluster center

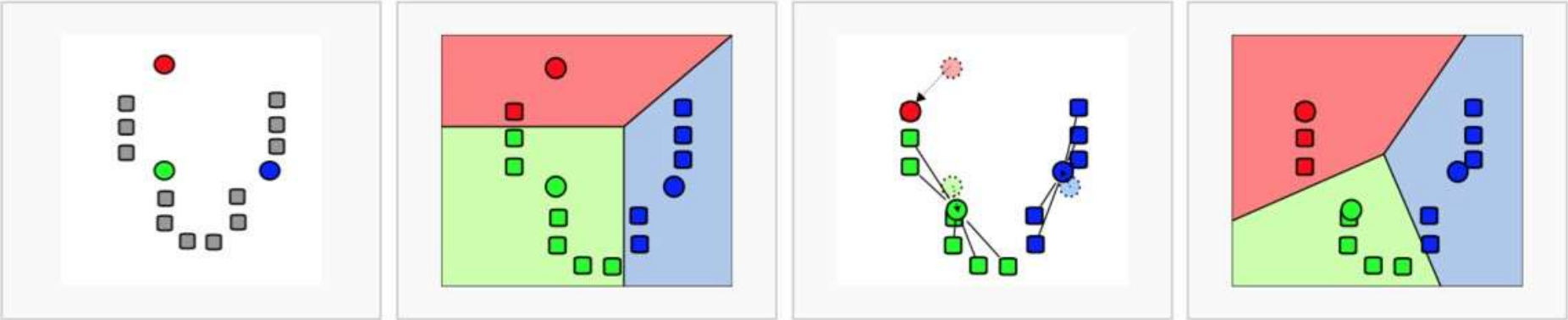
$$z_i \leftarrow \arg\min_j \|\mu_j - x_i\|_2^2$$

3. Revise cluster centers as mean of assigned observations

$$\mu_j = \frac{1}{n_j} \sum_{i:z_i=j} x_i$$

K-Means Algorithm

Demonstration of the standard algorithm



1. *k* initial "means" (in this case *k*=3) are randomly generated within the data domain (shown in color).

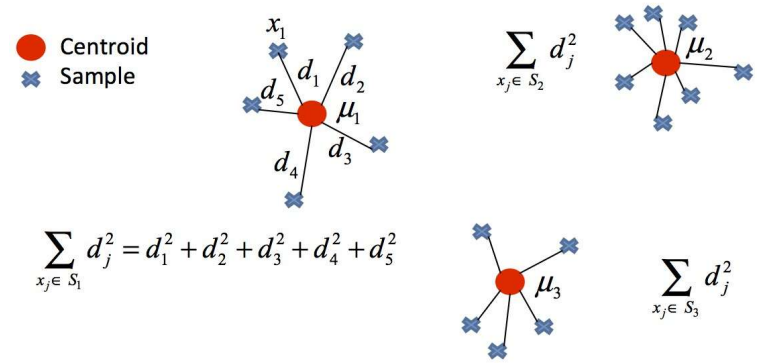
2. *k* clusters are created by associating every observation with the nearest mean. The partitions here represent the Voronoi diagram generated by the means.

3. The centroid of each of the *k* clusters becomes the new mean.

4. Steps 2 and 3 are repeated until convergence has been reached.

K Means Example

1. Pick seeds
2. Reassign clusters
3. Compute centroids
4. Reassign clusters
5. Compute centroids
6. Reassign clusters
7. Converged!



$$\min_S E(\mu_i) = \sum_{x_j \in S_1} d_j^2 + \sum_{x_j \in S_2} d_j^2 + \sum_{x_j \in S_3} d_j^2$$

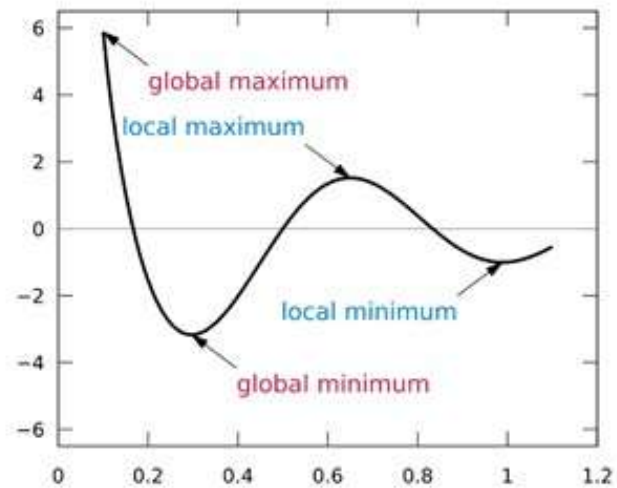
K-MEANS STRENGTHS & WEAKNESS

Strengths

- Compact and separate clusters
- Complexity is $O(nktd)$.
- Applied on Numeric data
- Relatively scalable
- Efficient in processing large datasets.

Weakness

- NP hard problem
- No guaranteed convergence to global optimum.
- Results depend on the initial random selection of cluster centers.



K-MEANS EXAMPLE

Use the k-means algorithm and Euclidean distance to cluster the following 5 examples into 2 clusters: $A = (0, 1)$; $B(3, 0)$; $C(2, 4)$; $D(2, 1)$; $E(3, 5)$

STEP 1 : Compute the Euclidean distance matrix

	A	B	C	D	E
A	0	3.16	3.61	2	5
B		0	4.12	1.414	5
C			0	3	1.414
D				0	4.12
E					0

STEP 2 : Randomly choose two cluster centroids.

Let A and C be the cluster centroids.

STEP 3 : Cluster 1: { A, B, D } and Cluster 2: { C, E}

K-MEANS EXAMPLE

STEP 4 : Compute the Centroids,

Cluster1 = mean of A,B,D = $\frac{0+3+2}{3}, \frac{1+0+1}{3} = (1.66, 0.66)$

Cluster 2 = mean of C,E = $\frac{2+3}{2}, \frac{4+5}{2} = (2.5, 4.5)$

STEP 5 : Repeat; Compute the Euclidean distance matrix to the cluster centers.

	A	B	C	D	E	C1	C2
A	0	3.16	3.61	2	5		
B		0	4.12	1.414	5		
C			0	3	1.414		
D				0	4.12		
E					0		
C1	1.69	1.49	3.35	0.48	4.54	0	
C2	4.30	4.52	0.70	3.53	0.70	3.93	0

K-MEANS EXAMPLE

STEP 6 : Cluster 1: { A, B, D }
Cluster 2: { C, E}

STEP 7 : Recompute the Centroids,
Cluster1 = mean of A,B,D = $\frac{0+3+2}{3}, \frac{1+0+1}{3} = (1.66, 0.66)$
Cluster 2 = mean of C,E = $\frac{2+3}{2}, \frac{4+5}{2} = (2.5, 4.5)$

STEP 8 : Stop as converged or no change is seen.

K-MEANS - INITIAL CENTROIDS

- Multiple runs – Helps, but probability is not favourable.
- Sample and use hierarchical clustering to determine initial centroids.
- Select more than k initial centroids and then select among these initial centroids, the most widely separated ones.
- Post-processing

Determining the Number of Clusters

1. Set the number of clusters to about $\sqrt{\frac{n}{2}}$ for a data set of n points.
2. Use elbow method.
   - Elbow method is based on the observation that increasing the number of clusters can help to reduce the sum of within-cluster variance of each cluster. A heuristic for selecting the right number of clusters is to use the turning point in the curve of the sum of within-cluster variances with respect to the number of clusters.
3. Use cross validation technique.
   - Compare the overall quality measure with respect to different values of k , and find the number of clusters that best fits the data.

Elbow Method - Numerical Problem

For the following dataset, using different values of $k = 2; 3; 4; 5; 6$ determine the best values of k using Elbow method.

$$D = \{2; 10; 12; 4; 25; 3; 30; 20; 11\}$$

Assumptions: Manhattan distance measure.

k=2: Assume initial centroids as 3 and 10.

$$C_1 = \{3; 2; 4\}$$

$$C_2 = \{10; 12; 25; 30; 20; 11\}$$

$$\begin{aligned} Variance &= (3-3)^2 + (3-2)^2 + (3-4)^2 + (10-10)^2 + (12-10)^2 \\ &\quad + (11-10)^2 + (20-10)^2 + (25-10)^2 + (30-10)^2 \\ &= 732 \end{aligned}$$

Elbow Method - Numerical Problem

k=3: Assume initial centroids as 3 ; 10 and 25.

$$C_1 = \{3; 2; 4\}$$
$$C_2 = \{10; 12; 11\}$$
$$C_3 = \{25; 30; 20\}$$
$$Var = (3-3)^2 + (3-2)^2 + (3-4)^2 + (10-10)^2 + (12-10)^2 + (11-10)^2 + (20-25)^2 + (25-25)^2 + (30-25)^2 = 57$$

Elbow Method - Numerical Problem

k=4: Assume initial centroids as 3 ; 10; 20 and 30.

$C_1 = \{3; 2; 4\}$

$C_2 = \{10; 12; 11\}$

$C_3 = \{25; 20\}$

$C_4 = \{30\}$

$$Var = (3-3)^2 + (3-2)^2 + (3-4)^2 + (10-10)^2 + (12-10)^2 + (11-10)^2 + (20-25)^2 + (25-25)^2 + (30-30)^2 = 32$$

Elbow Method - Numerical Problem

k=5: Assume initial centroids as 3 ; 10; 12; 20 and 25.

$C_1 = \{3; 2; 4\}$

$C_2 = \{10\}$

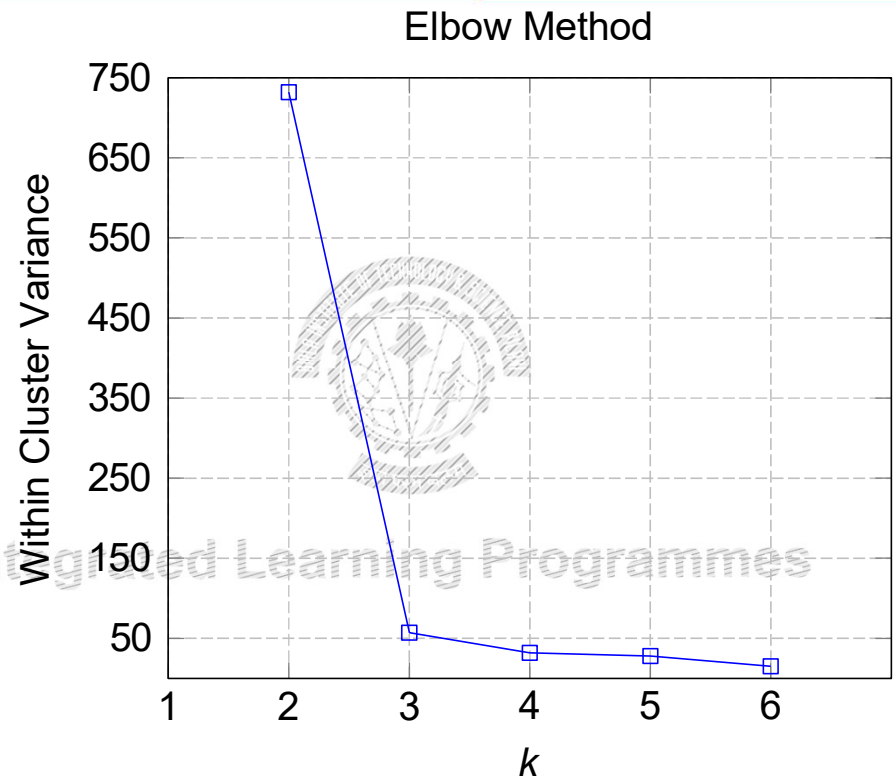
$C_3 = \{12; 11\}$

$C_4 = \{20\}$

$C_5 = \{25; 30\}$

$$Var = (3-3)^2 + (3-2)^2 + (3-4)^2 + (10-10)^2 + (12-12)^2 + (11-12)^2 + (20-20)^2 + (25-25)^2 + (30-25)^2 = 28$$

Elbow Method - Numerical Problem



Using the elbow method, $k = 3$ will yield better clustering results.

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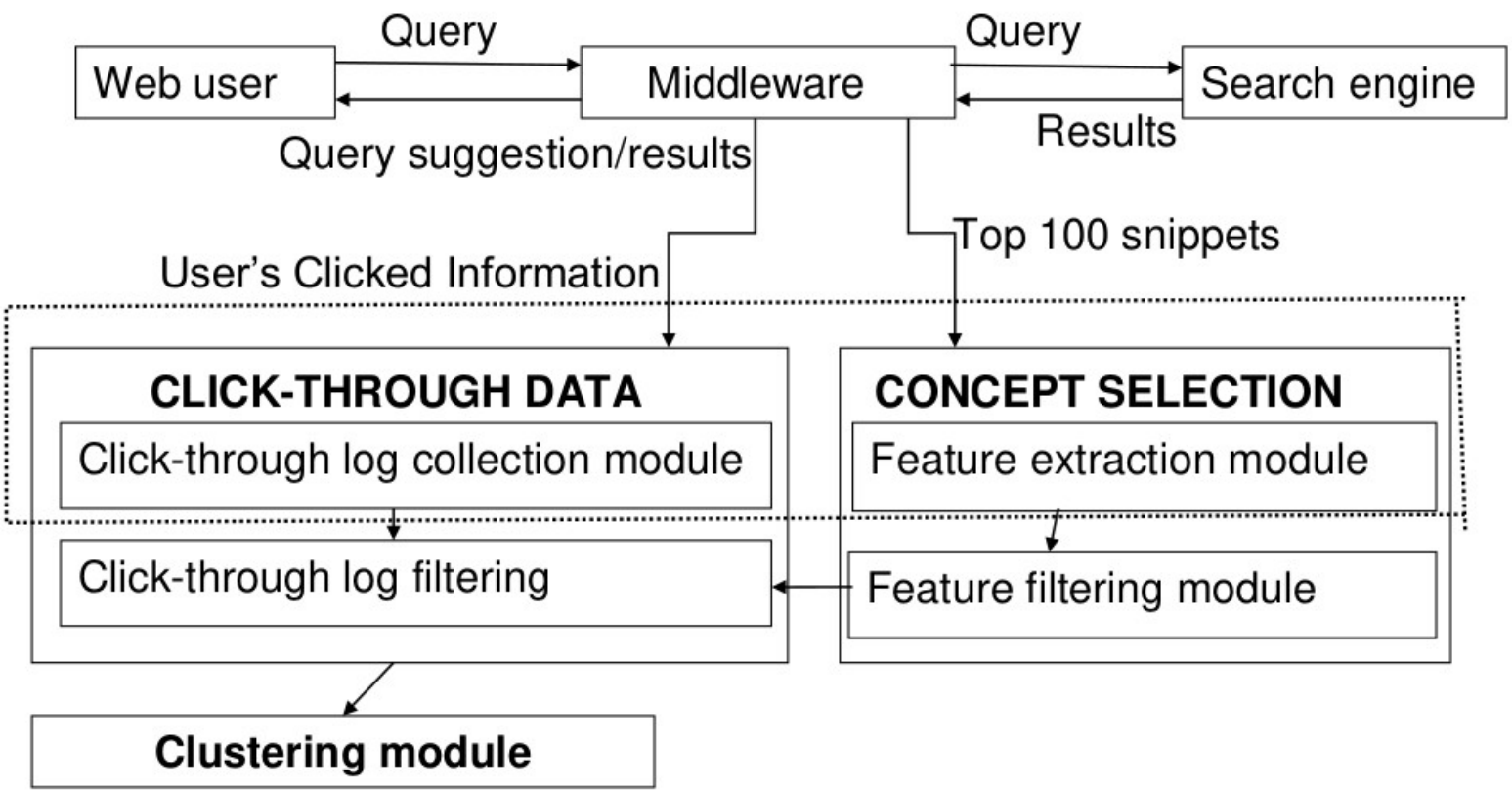
Clustering Internet Entities

- Group together similar Internet entities
  - Search queries {"rare books", "magazines"}
  - URLs {http://www.amazon.com, http://www.bn.com}
  - Users {Doug Beeferman, Adam Berger}
- Applications
  - Recommendation systems
  - Automatic taxonomies
  - Automatic communities

CLUSTERING APPROACH



Web Search Queries Coherence

- Categorization of web-search queries in semantically coherent topics is a crucial task
- The intent that drives users to submit their requests is an important element for meaningful aggregation of queries
- Activity fragments express a coherence among queries or among users' intent on the basis of their topical affinity

Issues in the Existing Approaches

Salton and McGill 1986:

- Considers only query terms for finding similar queries
- Unable to distinguish two queries of different meaning with common terms
- For example, ”Apple computers” and ”Apple Fruit”.

Beeferman and Berger 2000; Wen et al. 2001, 2002:

- Clusters queries based on the common clicks on the same pages.
- But clicking on the same page even for the similar queries is infrequent.
- Search engine gives different results in different ranking order for different queries at different times.

Issues in the Existing Approaches

Leung et al. 2008:

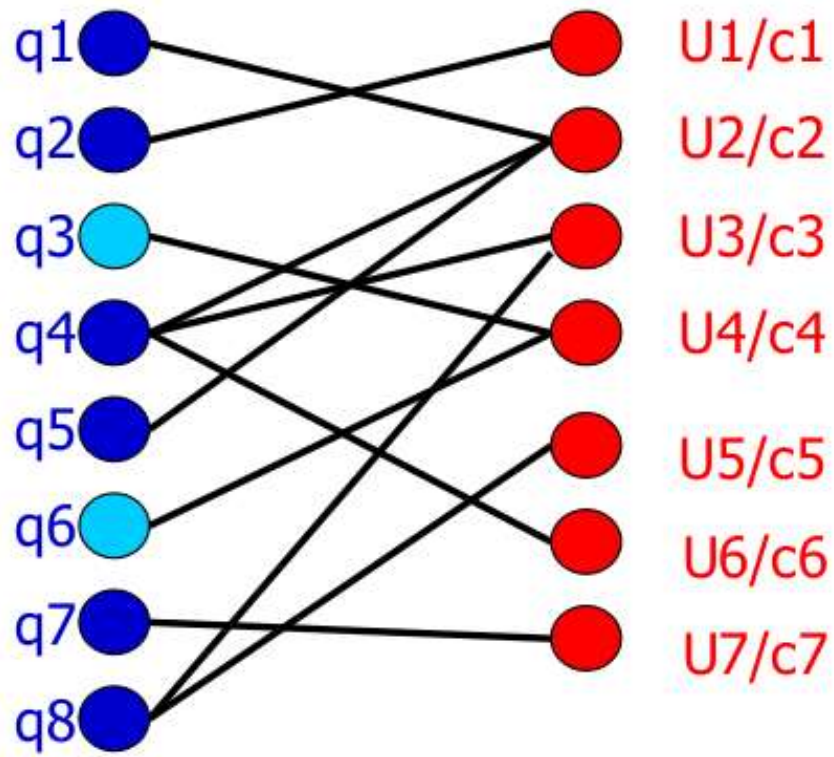
- Considers query and concept for query clustering, missing the following information:
  - Number of documents involved in relating the queries to the concepts.
  - Concepts which relates to a query are contained in the same or different documents.
  - If single concept is related to two or more queries, it is not known whether the concepts relating to a query are contained in the same or different sets of documents.

Query-Document and Query-Concept Mapping

Query-Document (QD) and Query-Concept (QC)

Steps:

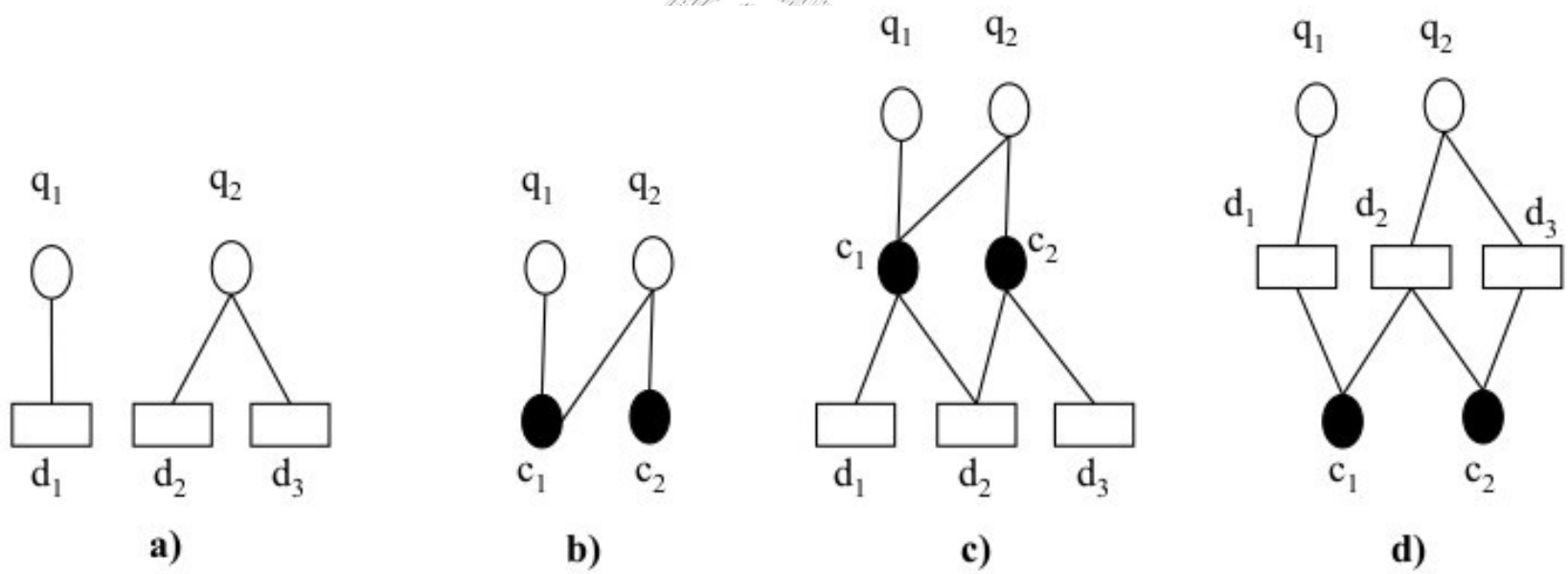
1. Bipartite Graph construction
2. Agglomerative Hierarchical Clustering Algorithm



Tripartite Graph Structure

Example 1:

Consider Figure , in which for queries q_1 and q_2, documents $\{d_1\}$ and $\{d_2, d_3\}$ are clicked respectively. The documents d_1, d_2 and d_3 contain the concepts c_1, c_1 & c_2 and c_2 respectively.



Tripartite Graph Structure

Example 2:

Consider queries ”eye parts” and ”Apple computer parts”. The ”Eyeball” concept (c_1) is present in snippets of document d_1 and d_2 which are clicked for q_1 and q_2 respectively.

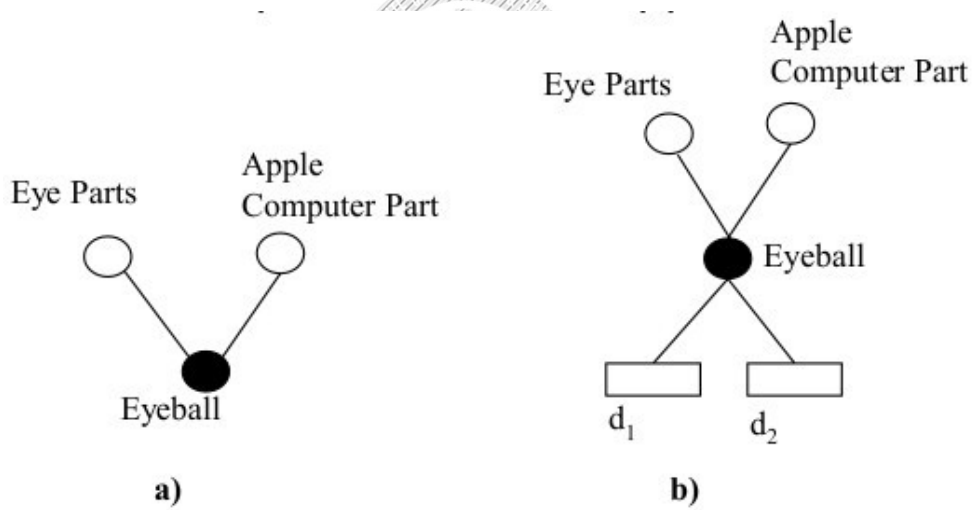


Figure: (a) & (b) show QC and QCD Graphs for Example 2 respectively

Tripartite Graph Structure

Example 3:
Consider a query ”Apple” and concepts, ”iPod” and ”iPhone”, which are present in the snippet of document d_1 clicked by the user

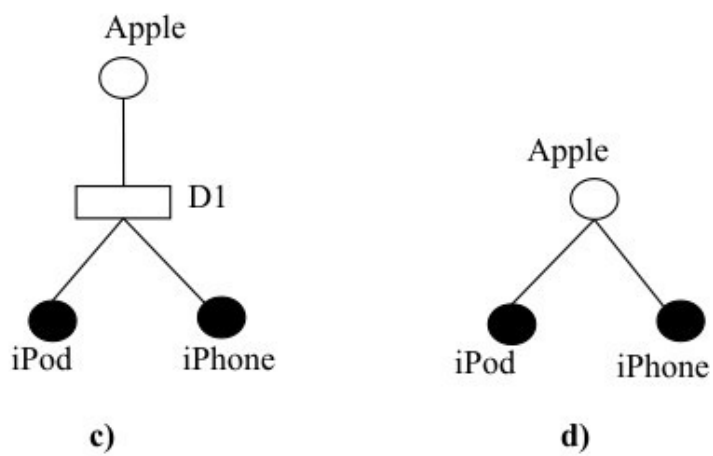


Figure: (c) & (d) show QDC and QC Graphs for Example 3 respectively

Clustering Module

The QDC-Clustering comprises of the following steps:

1. STEP1: QDC tripartite graph Construction
   - Left side vertices correspond to queries and right side vertices correspond to concepts and vertices between them correspond to documents.
2. STEP2: An agglomerative clustering algorithm is applied on the QDC tripartite graph constructed in step 1
3. STEP3: The clusters of queries and related clusters of documents and related clusters of concepts are extracted from the output of step 2.

References

1. Chapter 16
   C. D. Manning, P. Raghavan and H. Schutze. Introduction to Information Retrieval. 2008. http://nlp.stanford.edu/IR-book/
2. https://proceedings.neurips.cc/paper/2017/file/2e1b24a664f5e9c18f407b2f9c73e821-Paper.pdf
3. https://machinelearningmastery.com/clustering-algorithms-with-python/
4. https://docs.ampligraph.org/en/1.1.0/tutorials/ClusteringAndClassificationWithEmbeddings.html
5. https://www.coursera.org/learn/ml-clustering-and-retrieval/home/welcome

Thank You!